

USB 3.0 Connectivity using the Cypress EZ-USB FX3 Controller

PLC2 FPGA Days
June 20, 2012
Stuttgart



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Enclustra GmbH
FPGA Solution Center



- Enclustra Company Profile
- USB 3.0 Overview
 - What is new?
 - Why USB 3.0?
- EZ-USB FX3 Hardware and Software
 - Hardware
 - Software Framework
 - Memory Resources
 - GPIF 2 Designer Software
 - DMA Configurations
 - Debugging the FX3
 - Boot Options
- Slave Fifo Interface
 - Configuration
 - Threads
 - FPGA Implementation
 - Timings
 - Difficulties
- Enclustra USB 3.0 Solutions
 - FPGA Manager FX3
 - Mars PM3 Base Board
 - Performance Demo

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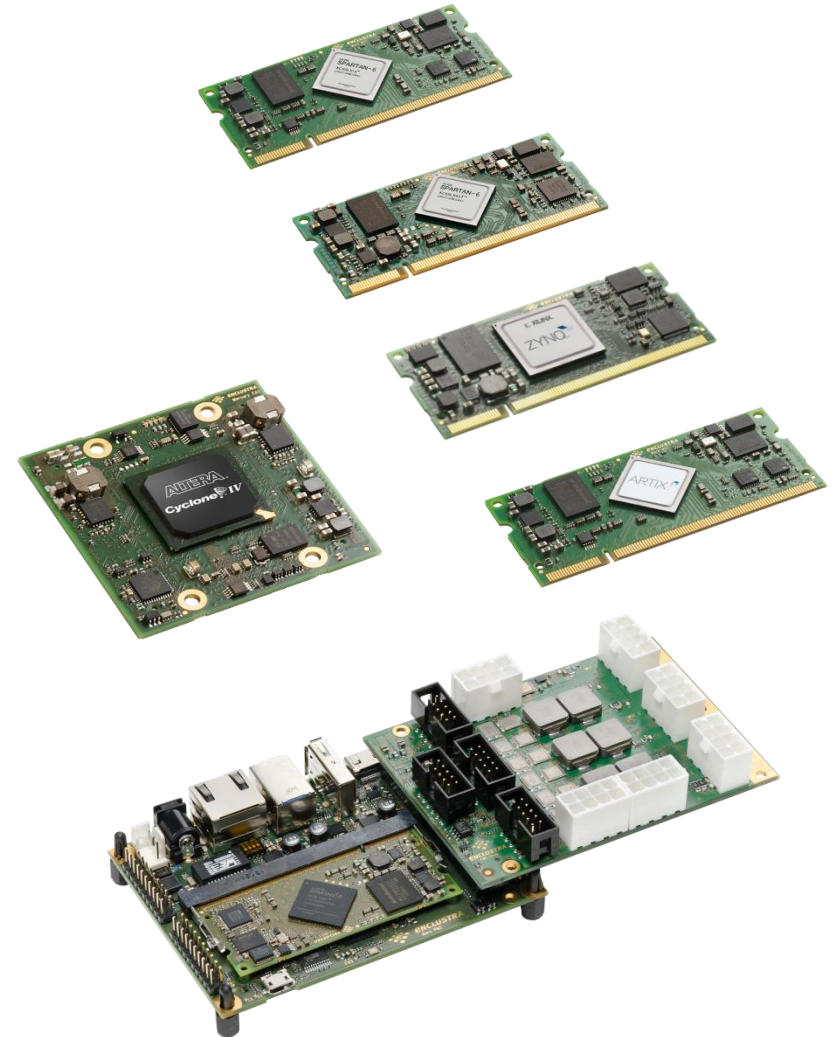
Enclustra Company Profile

FPGA Design Center

- Quick Facts
 - Founded in 2004
 - Located in the Technopark of Zurich
 - 8 FPGA Engineers, 1 Technician, 1 Secretary
 - Vendor-Independent
- FPGA Design Center
 - Provider of FPGA Design Services
 - HDL Firmware (VHDL, Verilog, C++)
 - Hardware (High-Speed, Analog, RF)
 - Embedded Software (for FPGA processors)



- Mars Family
 - SO-DIMM 67.6 x 30 mm
 - 108 User I/Os (Digital, LVDS, MGT, Analog)
 - 1-2 Ethernet Ports, 0-1 USB Port
 - 3.3V Single Supply Voltage
- Mercury Family
 - 56 x 54 mm
 - 146-168 User I/Os (Digital, LVDS)
 - 1 Ethernet Port, 1 USB Port
 - 5-15V Single Supply Voltage
- Base Boards & FMC Cards
- Custom „Dream“ Modules



- IP Cores

- Universal Drive Controller (DC/BLDC/SM)
- 2D-Accelerated Display Controller (LVCMOS/LVDS/DVI/HDMI)
- Camera Input Controllers (Camera Link/MIPI/LVDS)
- UDP Streaming Ethernet Controller (10/100/1G/10G)
- Resource-saving Memory Controllers (DDRx/QDRx/Flash)
- Available with AMBA AXI4 and Avalon-compliant interfaces



- IP Solutions

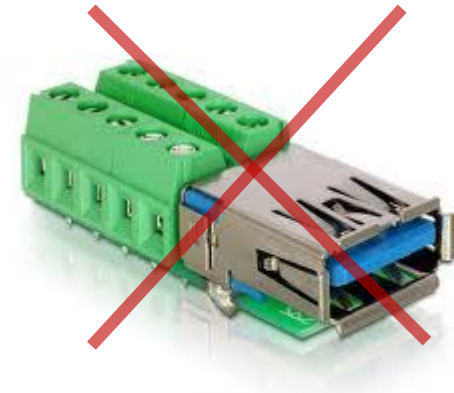
- FPGA Manager USB 3.0 Solution (Cypress EZ-USB FX3)
- Easy Licensing: Evaluation, Academic, Project, Site, Source
- Maintenance & Support
- Custom „Dream“ IP

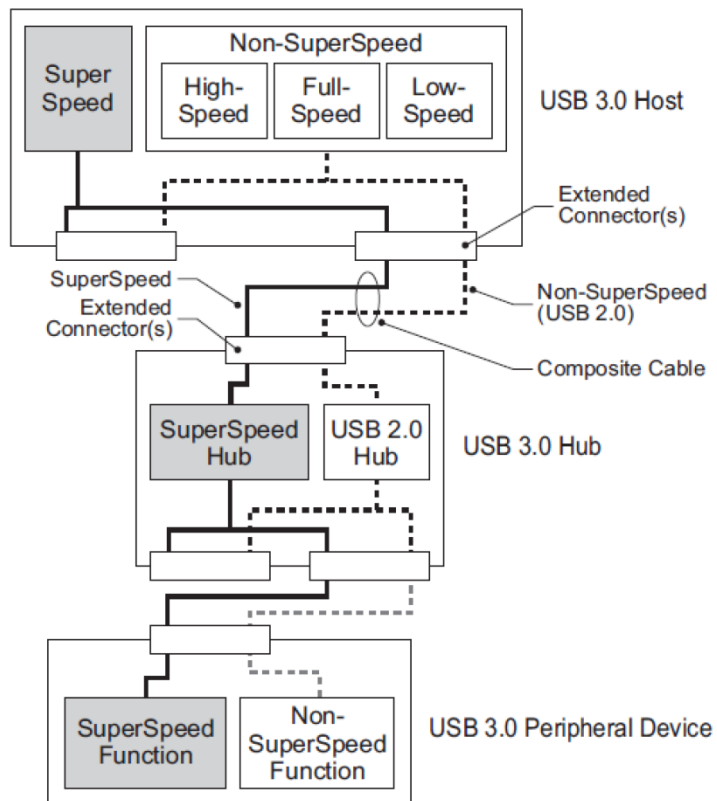


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Name	Data rate	Symbol rate
Low Speed	1.5 Mbit/s	1.875 Mbit/s
Full Speed	12 Mbit/s	15 Mbit/s
Hi-Speed	480 Mbit/s	600 Mbit/s
SuperSpeed	4 Gbit/s (500 MB/s)	5 Gbit/s

- USB 3.0 is using high-speed serial signaling similar to PCIe and relies on proper cabling and PCB design!





Note: Simultaneous operation of SuperSpeed and non-SuperSpeed modes is not allowed for peripheral devices.

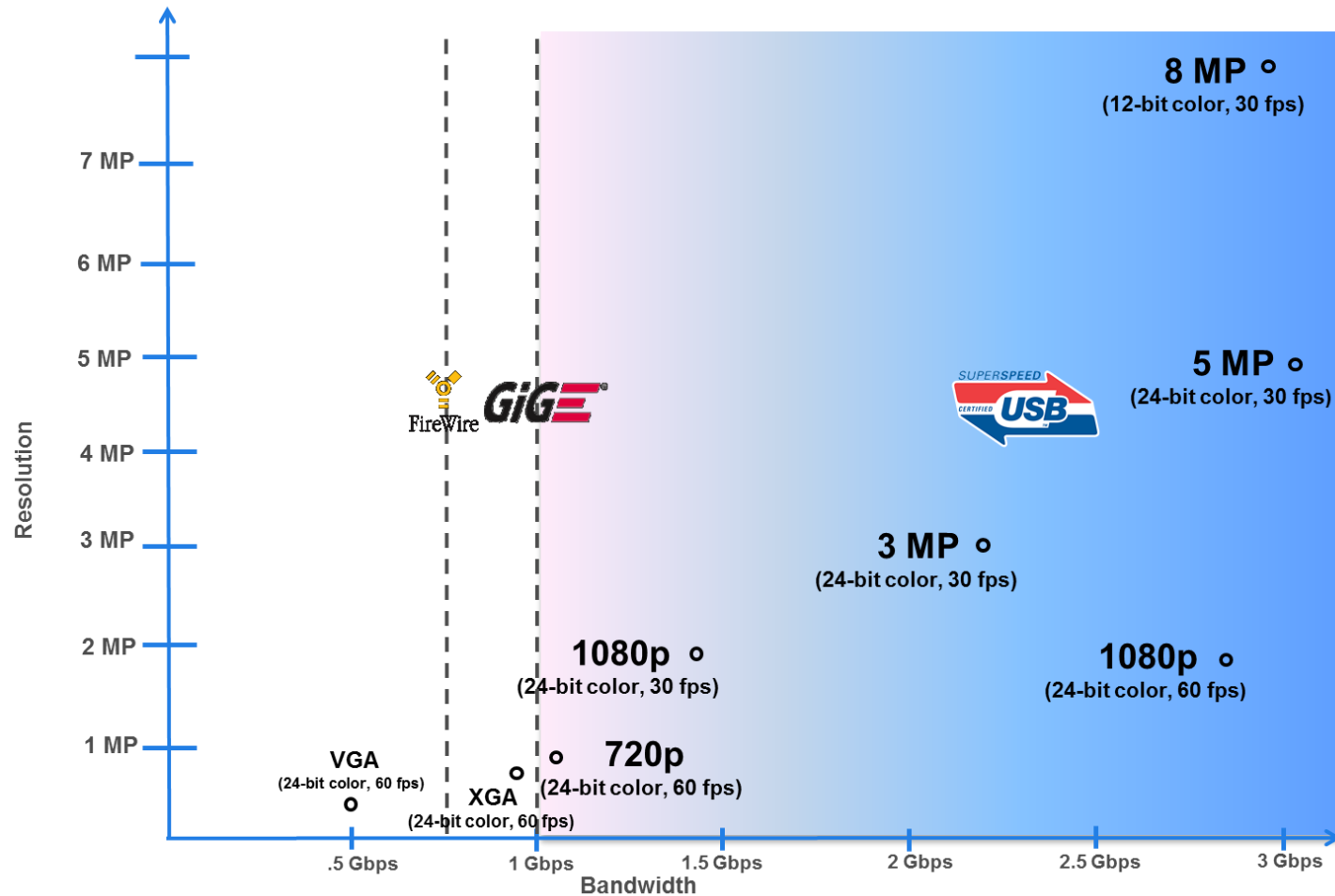
[1]

Pins Name	Description
VBUS	Power.
D-	USB 2.0 Differential Pair.
D+	
GND	Ground for Power Return.
SSRX-	SuperSpeed Receiver Differential Pair.
SSRX+	
SSTX-	SuperSpeed Transmit Differential Pair.
SSTX+	
GND_DRAIN	Ground for Signal Return.

[1]

- With USB 3.0 two additional differential pairs are introduced that support unidirectional data transfer at 5 Gbit/s
- New cables are therefore needed for USB 3.0
- USB 2.0 cables and most plugs are compatible but force USB 3.0 devices and hosts to work in USB 2.0 mode

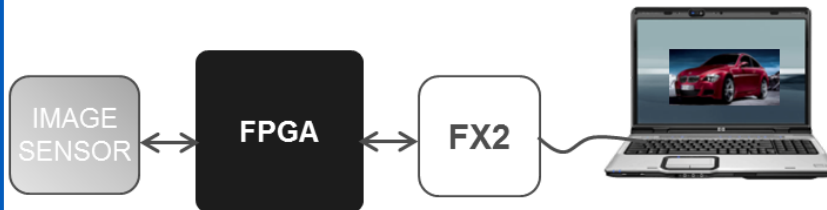
Overview - Why USB 3.0?



[2]

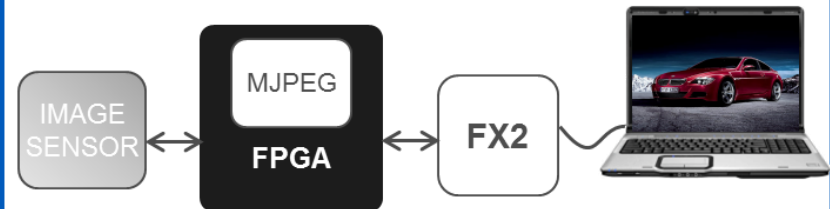
Overview - Why USB 3.0?

USB 2.0, NO COMPRESSION



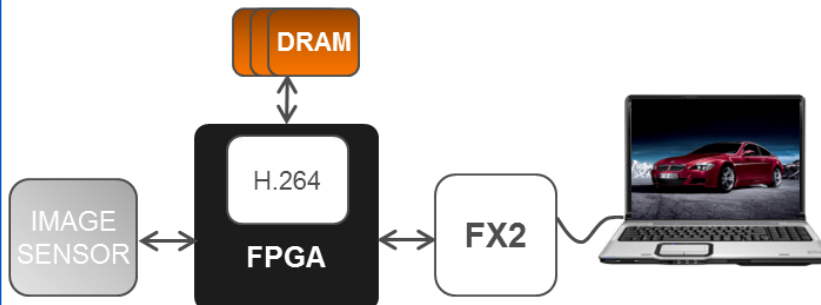
- **LIMITED VIDEO QUALITY, LOW FRAME RATE**

USB 2.0, MJPEG COMPRESSION



- **LOW COST IMPLEMENTATION**
- **GOOD ENOUGH IMAGE QUALITY BUT LOSSY**

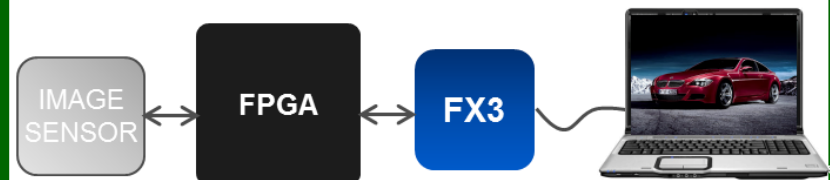
USB 2.0, H.264 COMPRESSION



- **HIGHER BOM COST AND MORE BOARD SPACE**
- **BETTER IMAGE QUALITY FOR MOST APPLICATIONS**



USB 3.0, NO COMPRESSION



- **LOWER BOM AND SMALLER BOARD**
- **HIGHEST IMAGE QUALITY**

[2]

Overview – Why USB 3.0?

INDUSTRIAL CAMERAS



GAMING



CONSUMER CAMERAS



MEDICAL IMAGING



SURVEILLANCE



VIDEO CONFERENCING



SCANNERS



BIOMETRICS



HD VIDEO CAPTURE



DATA ACQUISITION

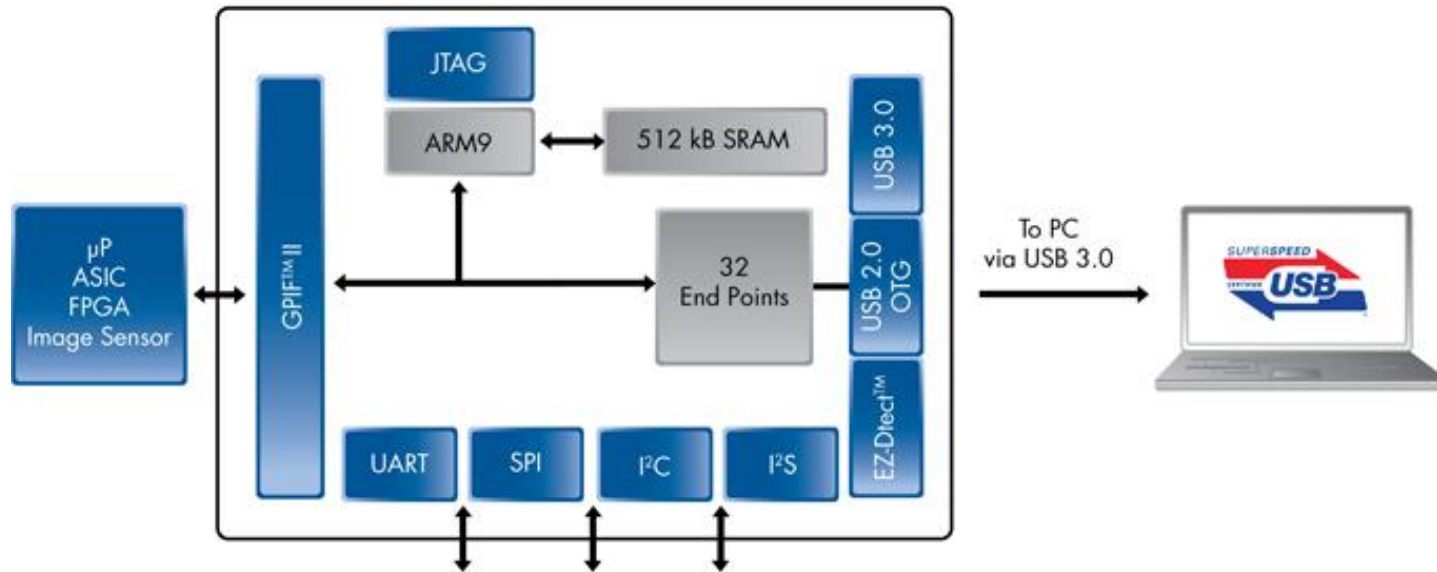


AUDIO MIXER



[2]

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FX3 Block Diagram [3]

- General Programmable Interface II (GPIF II) allows maximum flexibility for connecting data sources / sinks
- Automatic data transfer between USB and GPIF II port thanks to DMA engine
- ARM9 core can pre-process the data before sending to the host
- UART, SPI, I2C and I2S interfaces are used to load the FX3 firmware and/or configure external peripherals
- USB Charger and accessory detection (EZ-Dtect)

- 32-bit, 200 MHz ARM926EJ-S core
- The core has direct access to 16 kB of Instruction Tightly Coupled Memory (TCM) and 8 kB of Data TCM
- JTAG interface for firmware download and debugging
- 512 kB of embedded SRAM for code and data
- 8 kB of instruction and data cache
- DMA connectivity between the various peripherals (i.e. USB, GPIF II, I2S, SPI, UART)
- Industry-standard development tools for ARM926EJ-S can be used

[AN76405 - EZ-USB® FX3 Boot Options](#)

[AN70193 - EZ-USB® FX3 SPI Boot Option](#)

[AN68914 - EZ-USB® FX3 I2C Boot Option](#)

[AN70707 - EZ-USB® FX3 Hardware Design Guidelines and Schematic Checklist](#)

[AN77960 - Introduction to EZ-USB® FX3's High-Speed USB Host Controller](#)

[AN76348 - Migrating from EZ-USB® FX2LP™ Based Design to EZ-USB FX3 Based Design](#)

[AN75432 - USB 3.0 EZ-USB® FX3™ Orientation](#)

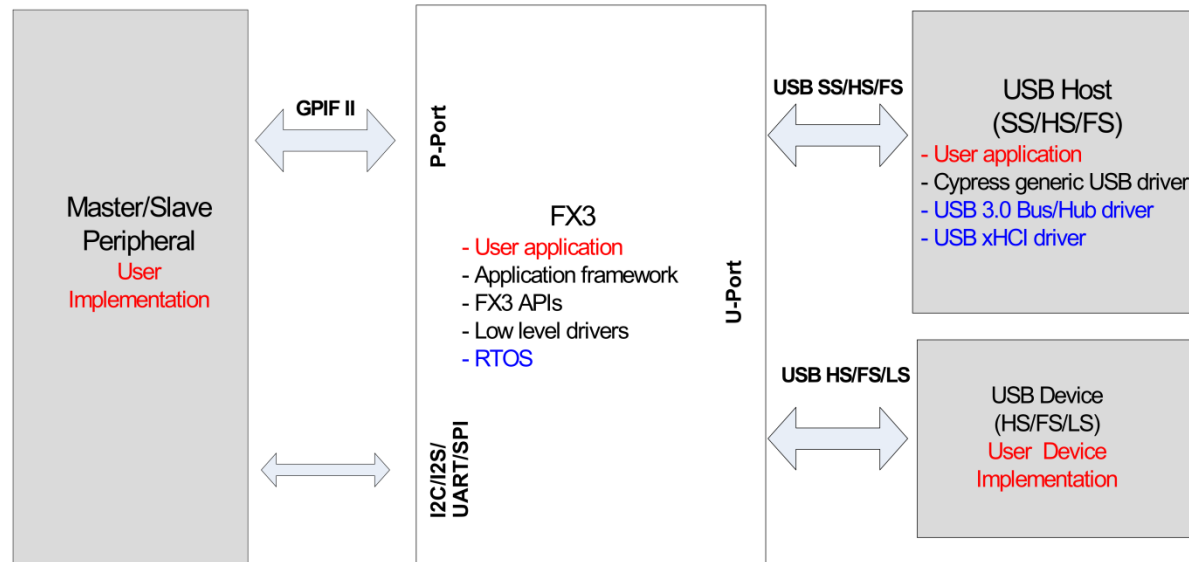
[AN75705 - Getting Started with FX3](#)

[AN68829 - Slave FIFO Interface for EZ-USB® FX3™: 5-Bit Address Mode](#)

[AN65974 - Designing with the EZ-USB® FX3 Slave FIFO Interface](#)

[AN73304 - Booting EZ-USB® FX3 over Synchronous ADMux Interfaces](#)

[AN73150 - Booting EZ-USB® FX3 over High-Speed USB](#)



Cypress provided software
 User or customer software
 Third-party or platform software

[4]

- Application framework for FX3 provided by Cypress
- Host USB library with simple read and write function exists
- ThreadX RTOS running on FX3
- GPIF II interface can be configured using GPIF II Designer software

Cypress Libraries

Host
USB Library

FX3 API
Library

Application Examples

Bulk Loop
(Host & FX3)

Streamer
(Host & FX3)

Control
Center
(Host only)

GPIF II
Configuration
Examples

Development Tools

Eclipse

GCC ARM
Compiler

GPIF II
Designer

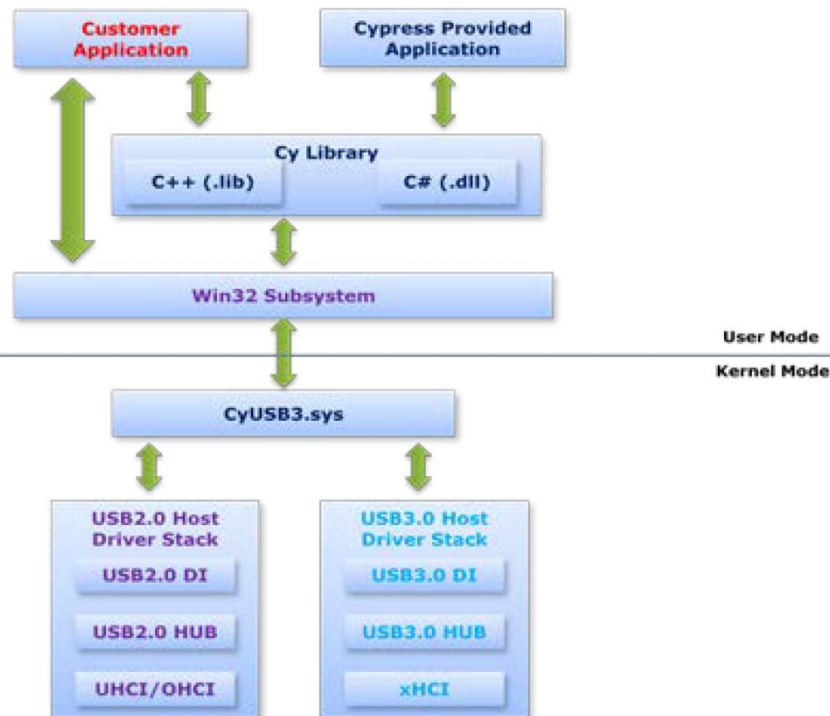
Others

USB Driver

Documen-
tation

- All necessary tools except compiler for host system are part of the SDK
- For on-chip debugging, an ARM9 compatible JTAG probe with GDB server is required
- Application can be downloaded to RAM and I2C / SPI attached memory over USB using the Control Center application

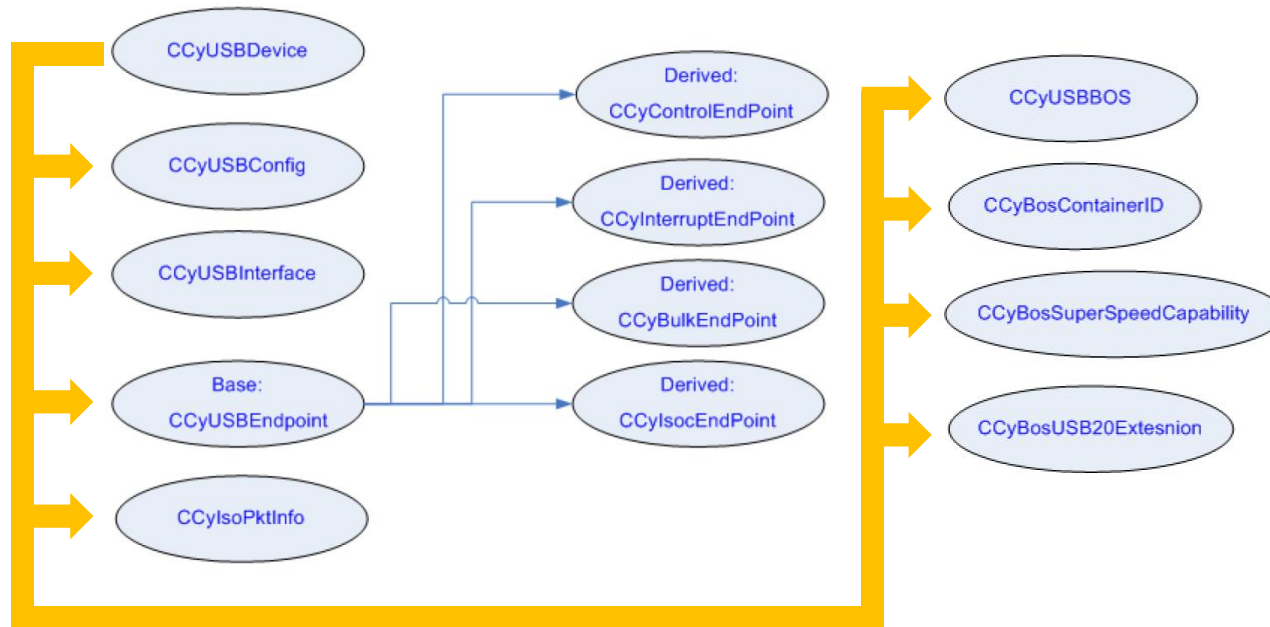
Software Framework – Host USB Library



Cypress provided software
 Customer software
 Third-party/Platform software
 Microsoft provided software

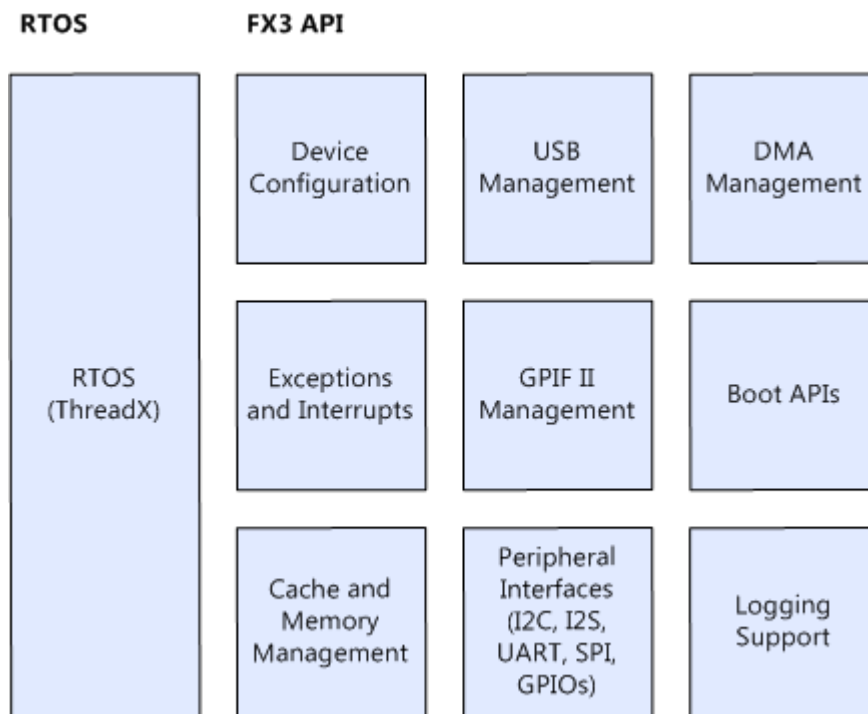
- C++ and C# library available
- Simple read and write functions to transfer data between host and device
- Cypress demo applications can be used as a starting point for the application development

[1]

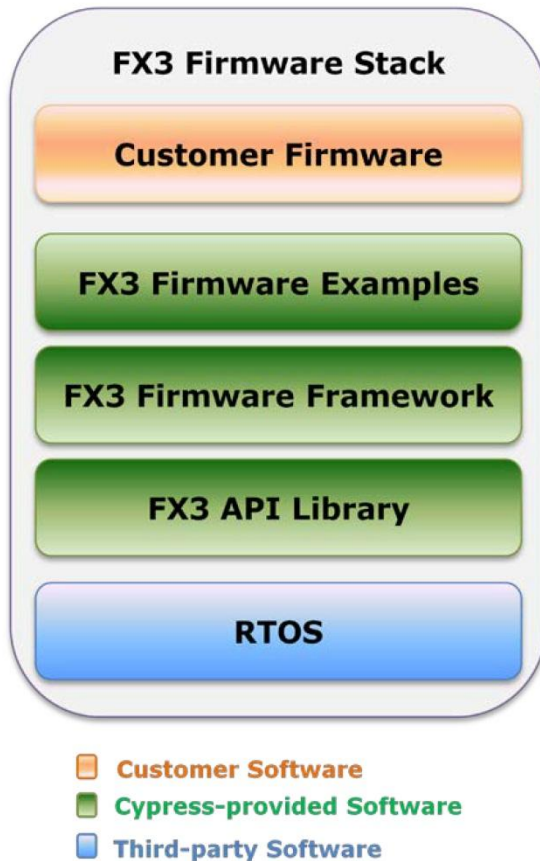


[5]

- The USB device is opened by instantiating the **CCyUSBDevice** class
- Based on the endpoint mode defined in the USB descriptor the appropriate derived **CCyUSBEndpoint** class is used

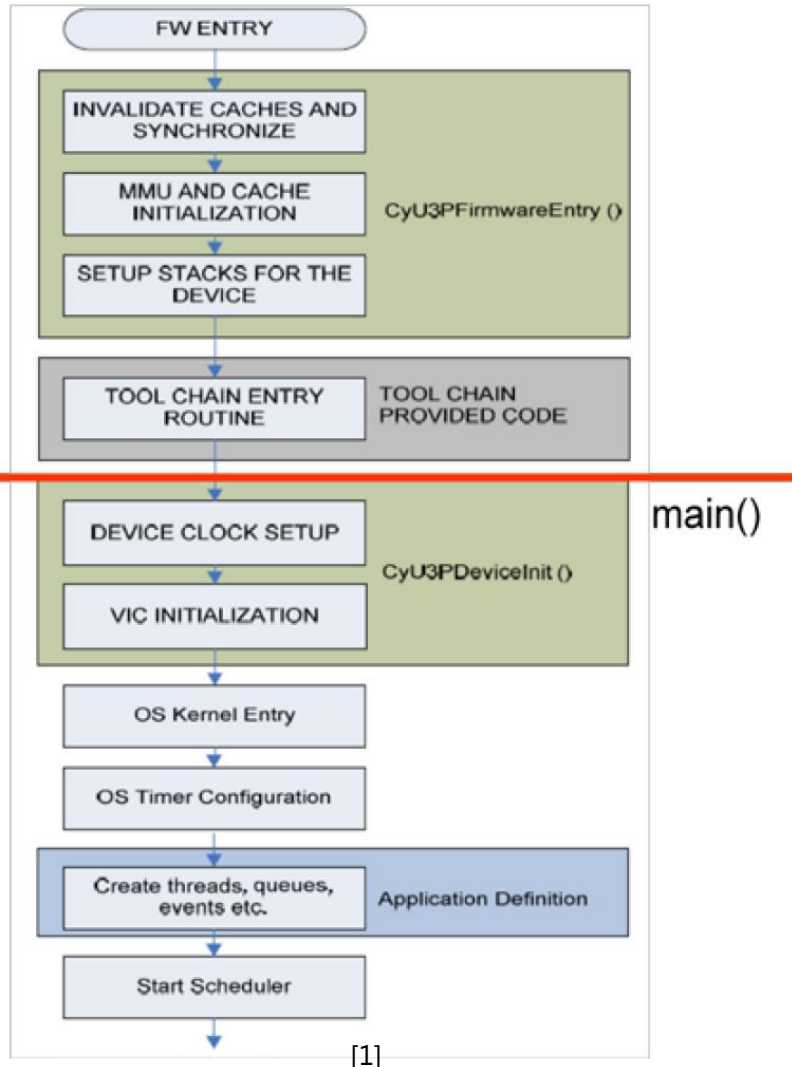


- ThreadX RTOS enables multi-threaded applications
- FX3 API provides a complete framework to configure and operate the FX3
- No direct access to FX3 registers and RAM necessary

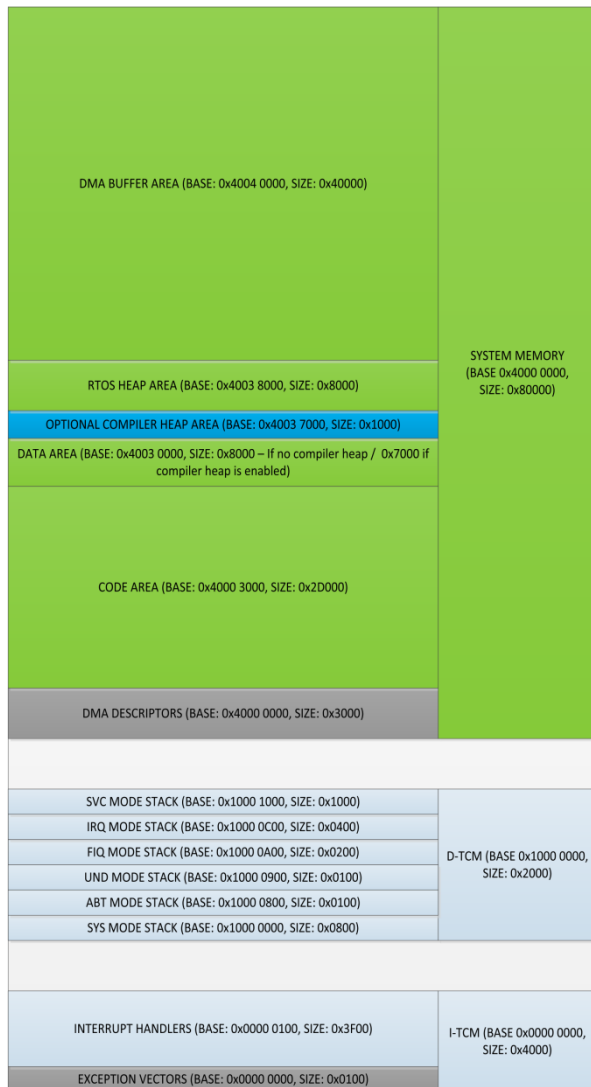


- ThreadX RTOS supports:
 - Threads
 - Message Queues
 - Semaphores
 - Mutex
 - Memory Allocation
 - Events
 - Timer
- Cypress application examples can be used as a starting point for the development

[1]



- CPU and memory initialization is done before entering the `main()` function
- GPIOs and peripherals can be configured in the `main()` function using the FX3 API
- User application is running in its own thread
- DMA channels are set up in the user application



[4]

512 kB RAM partitioned into:

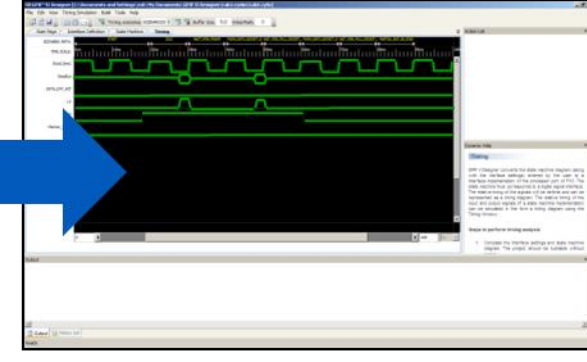
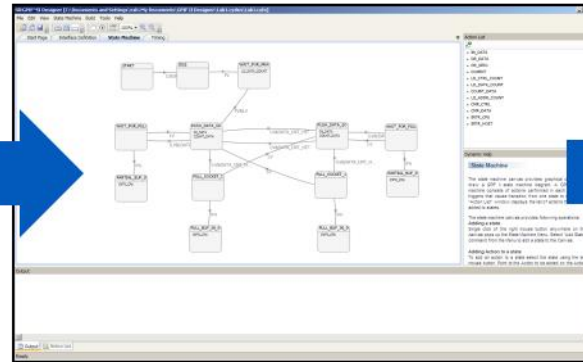
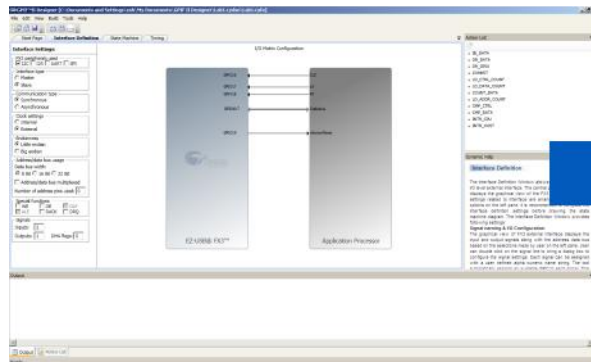
- 256 kB DMA buffer
- 32 kB User Data
- 180 kB Code
- 8 kB RTOS heap
- 8 kB Stack
- 4 kB Interrupt and exception vectors

➤ Cypress delivered application framework and RTOS results already in 145 kB code size

1

2

3



STEP 1: CHOOSE YOUR PINS

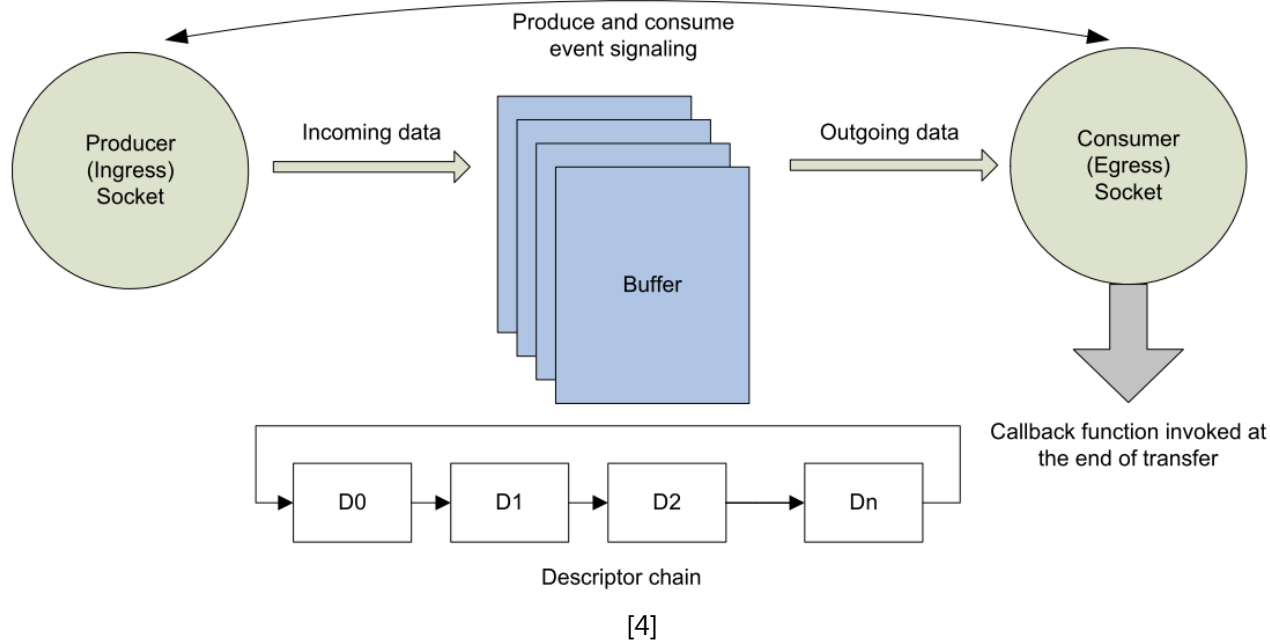
STEP 2: DESIGN YOUR INTERFACE

STEP 3: VERIFY THE TIMING!

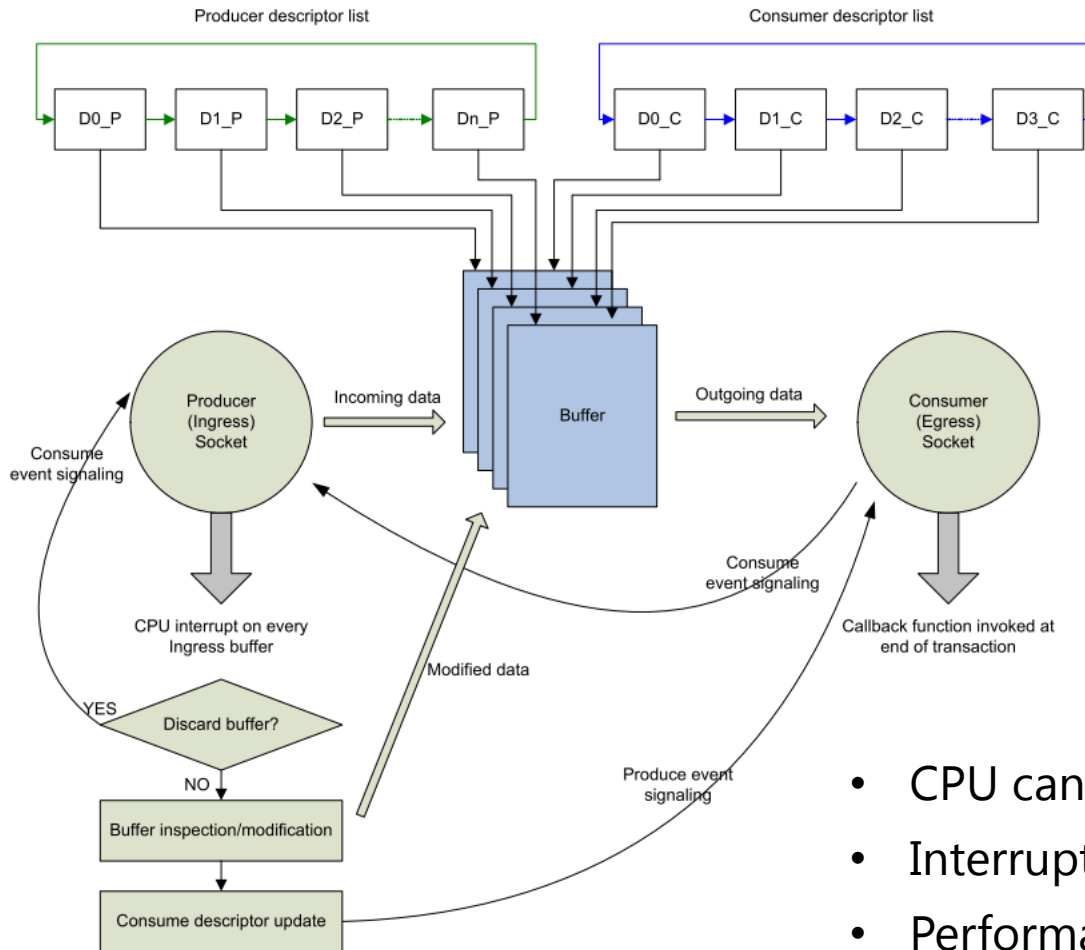
[2]

- Functions as master or slave
- Provides 256 firmware programmable states
- Supports 8 Bit, 16 Bit and 32 Bit parallel data bus
- Enables interface frequencies up to 100 MHz
- Supports 14 configurable control pins when 32 Bit data bus is used. All control pins can be either input/output or bidirectional.
- Supports 16 configurable control pins when 16 or 8 Bit data bus is used. All control pins can be either input/output or bidirectional.

DMA – Auto Channel

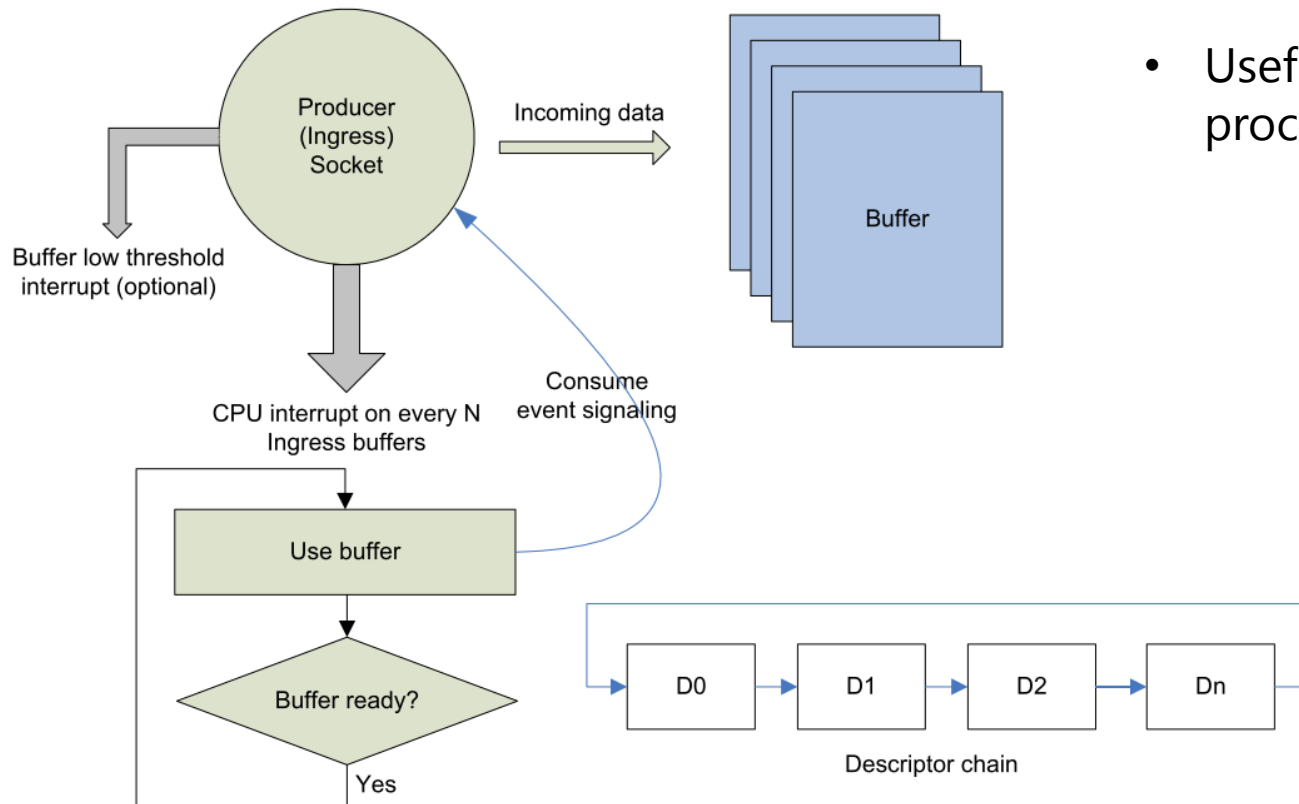


- No CPU intervention
- Data cannot be modified
- Maximum performance
- Optional call-back functionality



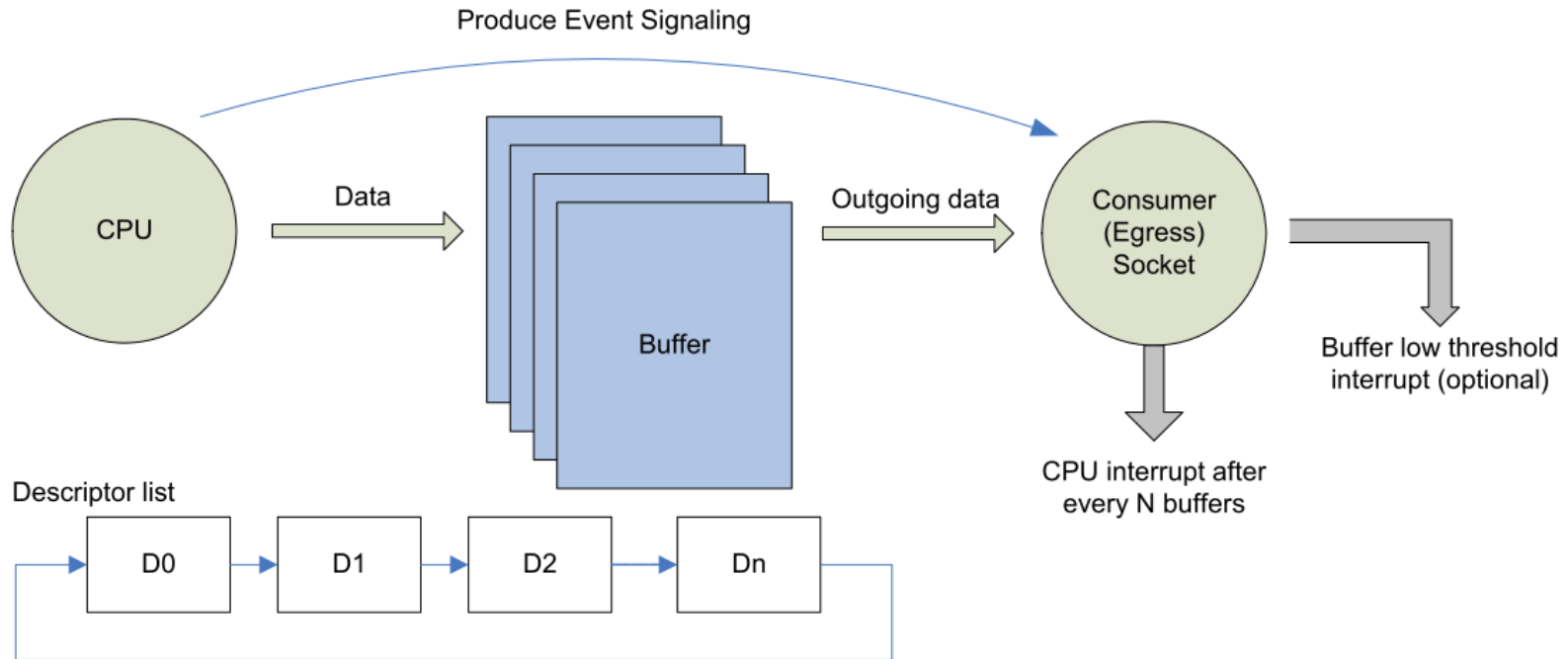
[4]

- CPU can modify every data packet
- Interrupt driven
- Performance severely reduced



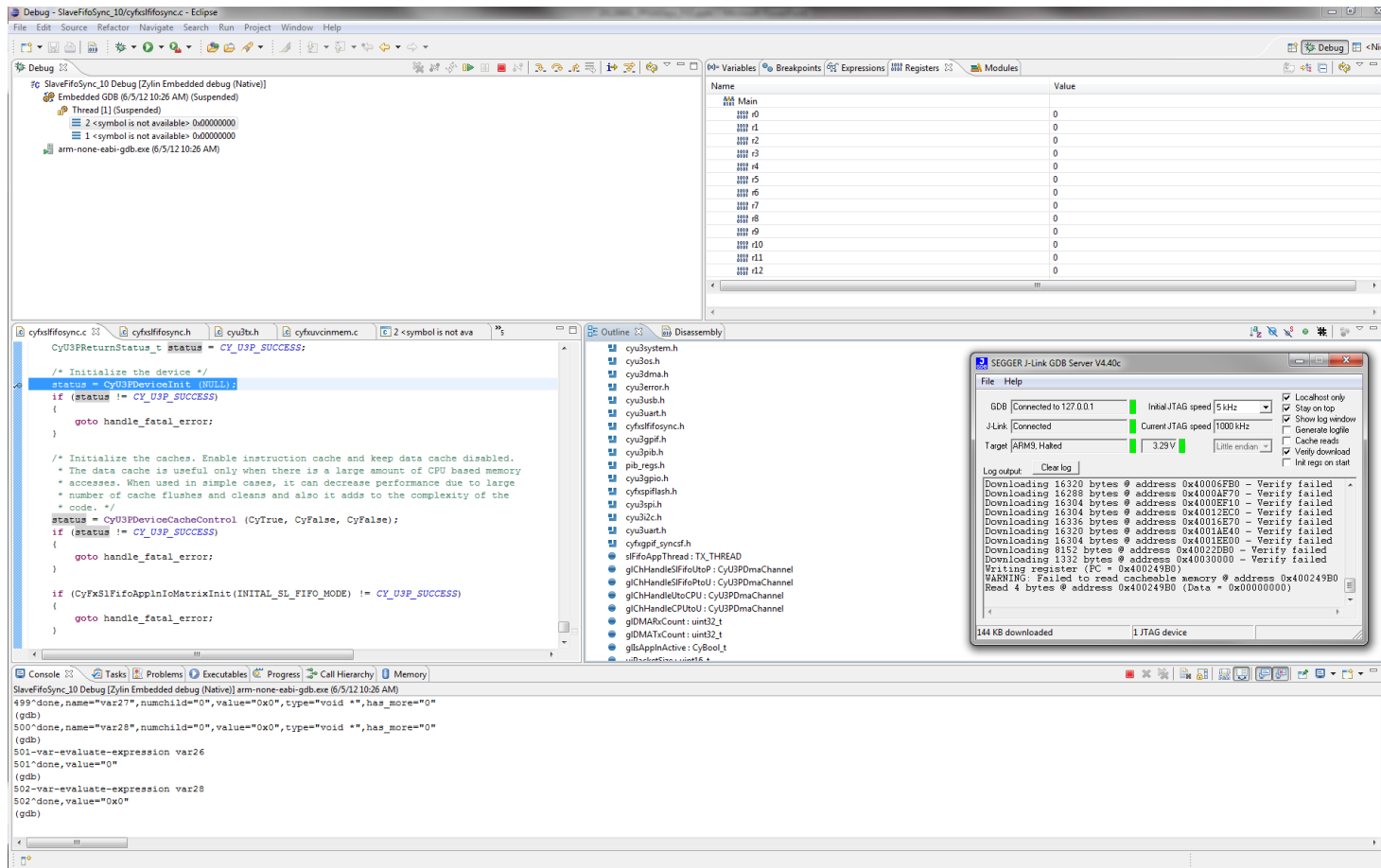
- Useful if data is read and processed on FX3

[4]



[4]

- One possible application is to send debug messages over UART



The screenshot shows the Eclipse IDE interface for debugging the FX3 application. The main window displays the 'Debug' console with the 'SlaveFifoSync_10/cyufalifosync' project. The 'Variables' panel shows the 'Main' thread with various registers (r0-r12) all set to 0. The 'Console' panel shows the output of the 'arm-none-eabi-gdb.exe' process, including the initialization of the device and caches. The 'SEGGER J-Link GDB Server V4.40c' window is also visible, showing the connection to the JTAG device and the download progress of the application code.

- FX3 application can be downloaded and debugged over JTAG (e.g. Segger or Signum JTAG probe) using Eclipse and GDB server

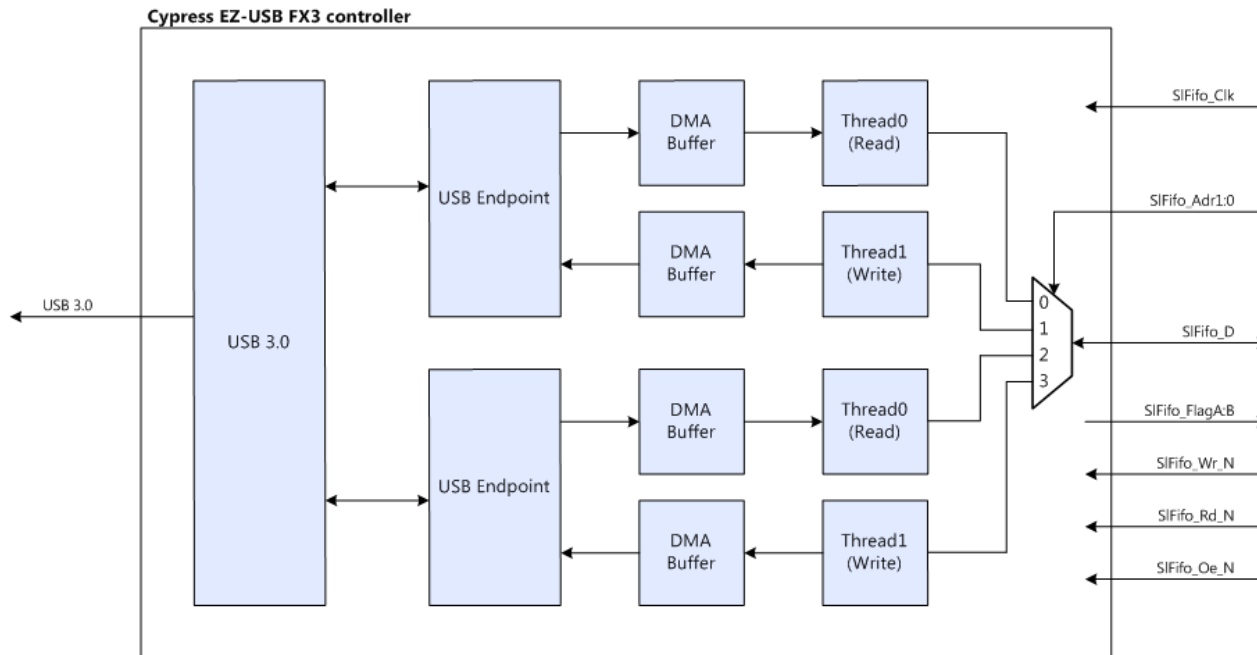
FX3 firmware can be loaded from various sources:

PMODE[2:0]^[2]	Boot From
F00	Sync ADMUX (16-bit)
F01	Async ADMUX (16-bit)
F11	USB boot
F0F	Async SRAM (16-bit)
F1F	I ² C, On Failure, USB Boot is Enabled
1FF	I ² C only
0F1	SPI, On Failure, USB Boot is Enabled

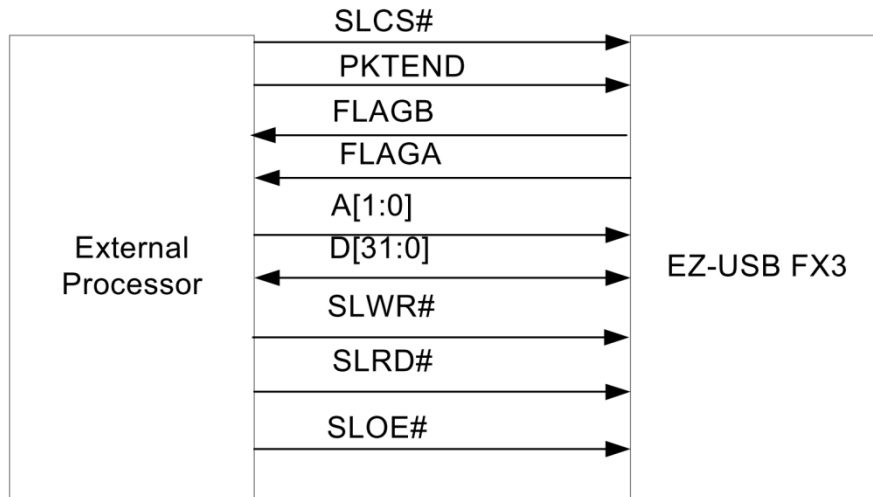
[3]

- When using USB boot mode, the firmware is downloaded by the host driver and therefore driver and firmware versions are always matching
- If the FX3 needs to work without USB connection, the firmware image can be stored using an external memory attached to the SPI or I2C interface

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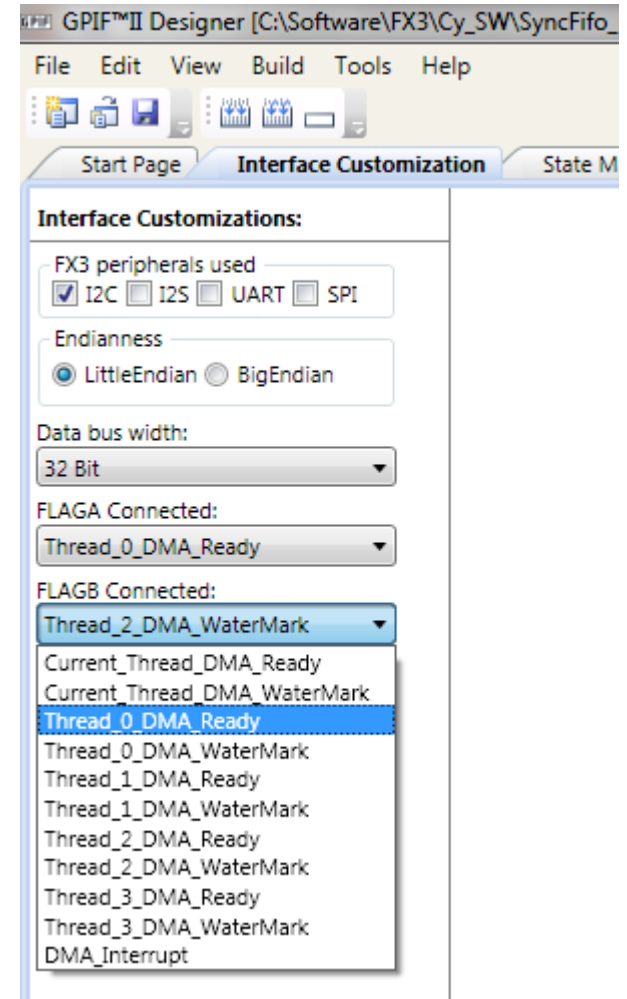


- The FX3 slave fifo interface provides four threads that are addressed using two address signals
- Each thread can be configured as read or write thread
- Flags can be assigned to a fixed thread or the currently addressed thread
- Each thread can be linked to a DMA channel and its associated buffer



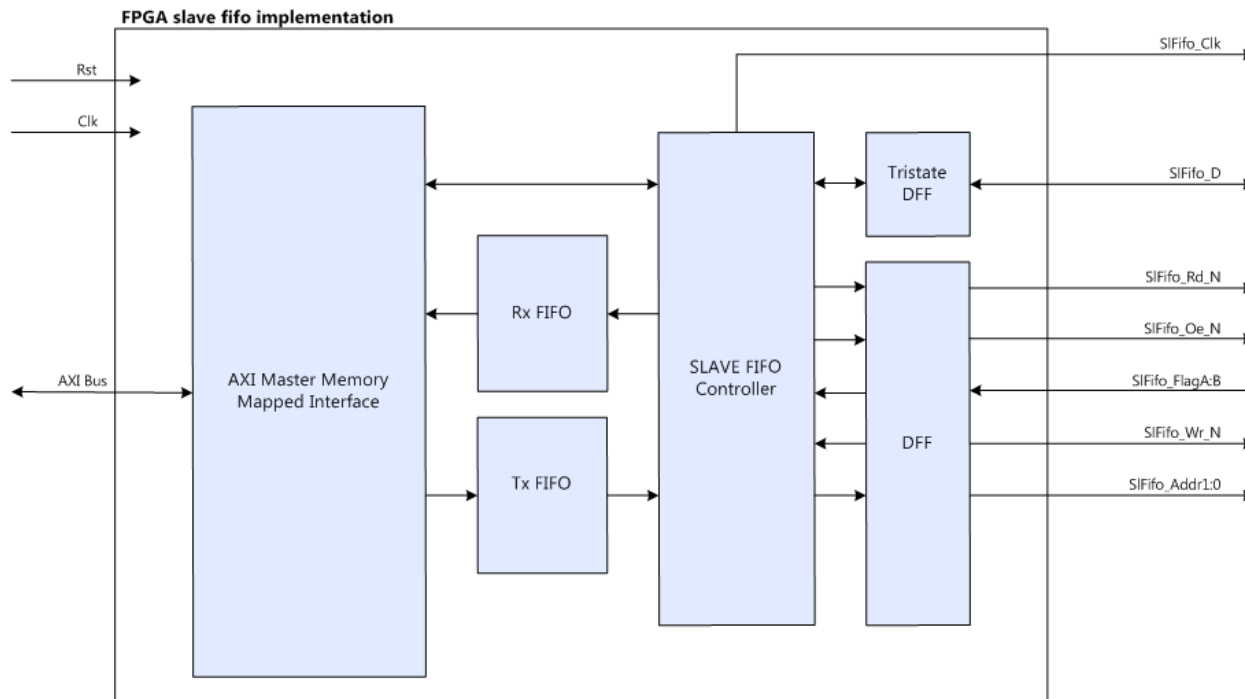
[3]

- FlagA and FlagB can be configured as full/empty and watermark flags
- Flags can be assigned to current thread or a fixed thread
- Data bus width can be set to 8, 16 or 32 Bit

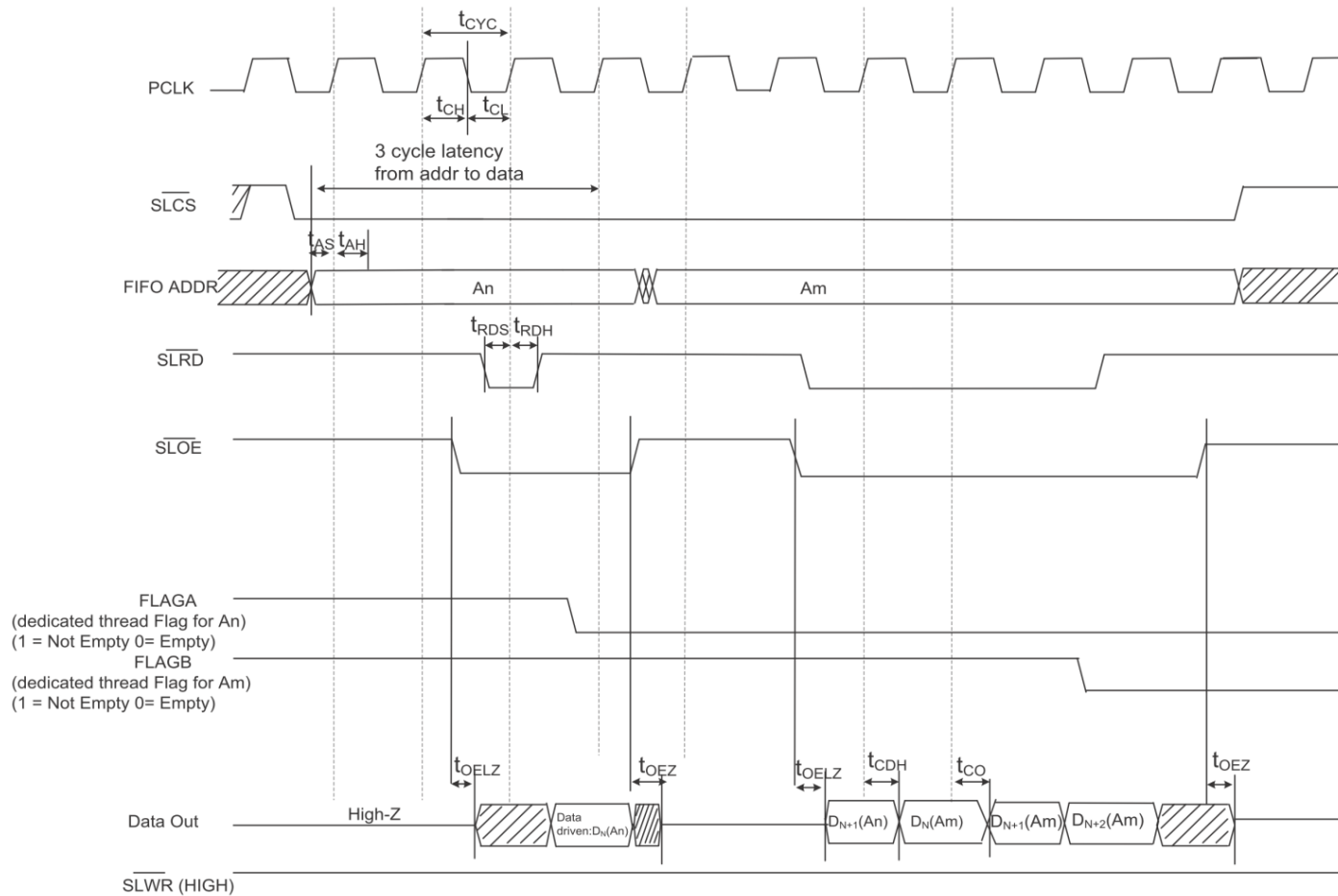


Slave Fifo Interface – FPGA Implementation

- Two threads are addressed using the *SlFifo_Addr* signal
- One thread is used as write and one as read thread
- Flag A is configured as read thread empty
- Flag B is configured as write thread almost full

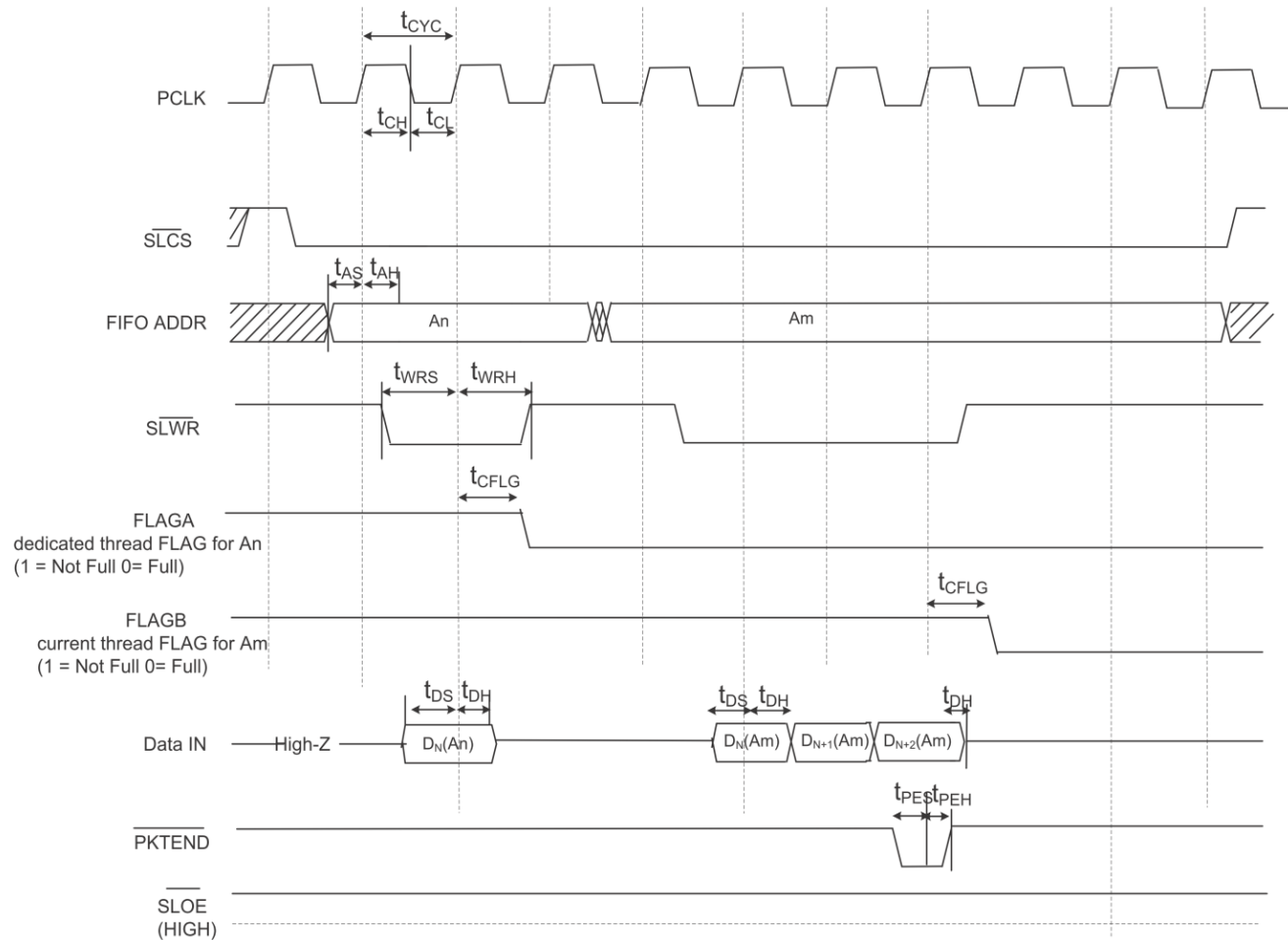


Synchronous Read Cycle Timing



[3]

Synchronous Write Cycle Timing



[3]

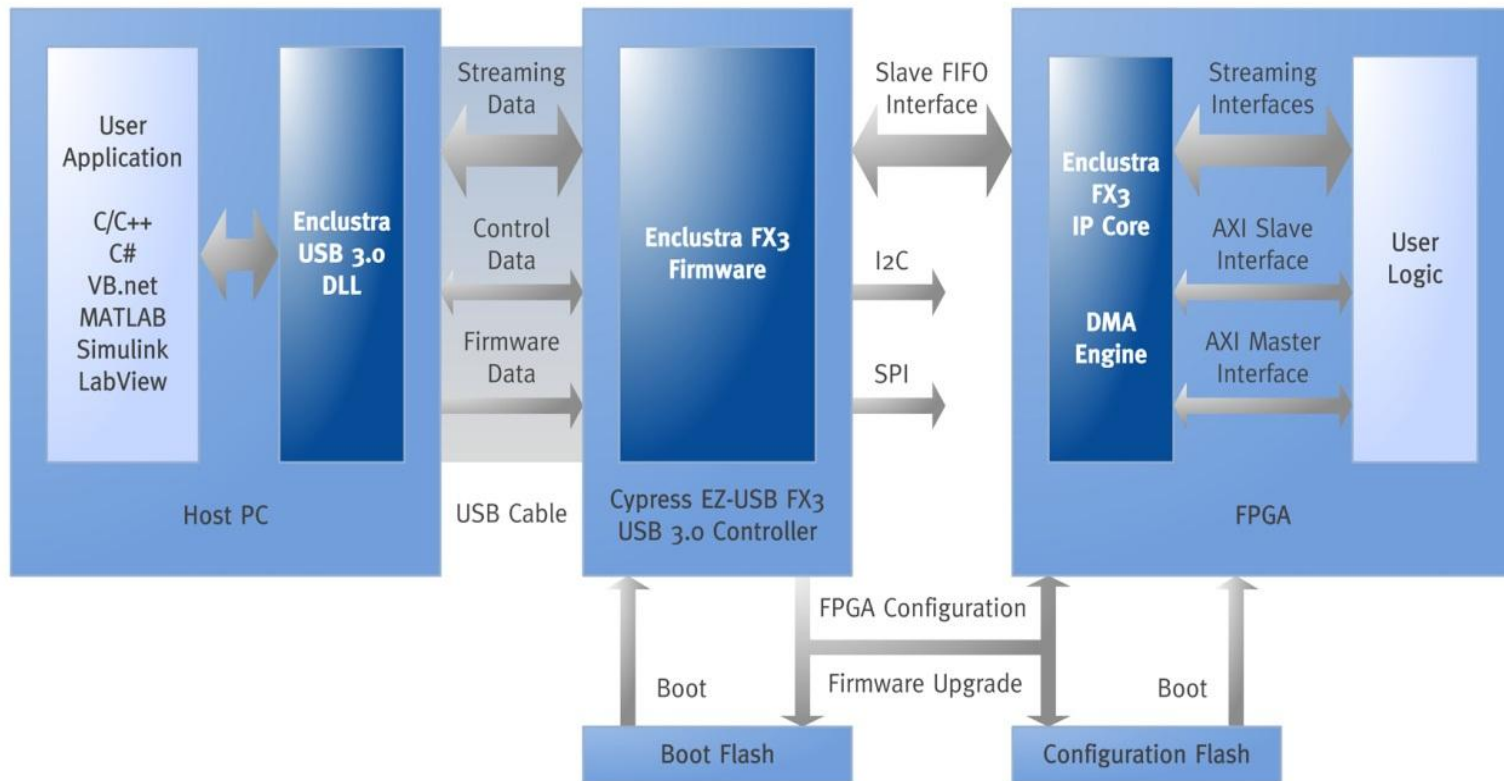
Parameter	Description	Min	Max	Units
FREQ	Interface clock frequency	–	100	MHz
tCYC	Clock period	10	–	ns
tCH	Clock high time	4	–	ns
tCL	Clock low time	4	–	ns
tRDS	SLRD# to CLK setup time	2	–	ns
tRDH	SLRD# to CLK hold time	0.5	–	ns
tWRS	SLWR# to CLK setup time	2	–	ns
tWRH	SLWR# to CLK hold time	0.5	–	ns
tCO	Clock to valid data	–	8	ns
tDS	Data input setup time	2	–	ns
tDH	CLK to data input hold	0.5	–	ns
tAS	Address to CLK setup time	2	–	ns
tAH	CLK to address hold time	0.5	–	ns
tOELZ	SLOE# to data low-Z	0	–	ns
tCFLG	CLK to flag output propagation delay	–	8	ns
tOEZ	SLOE# deassert to Data Hi Z	–	8	ns
tPES	PKTEND# to CLK setup	2	–	ns
tPEH	CLK to PKTEND# hold	0.5	–	
tCDH	CLK to data output hold	2	–	ns
Note Three-cycle latency from ADDR to DATA/FLAGS				

[3]

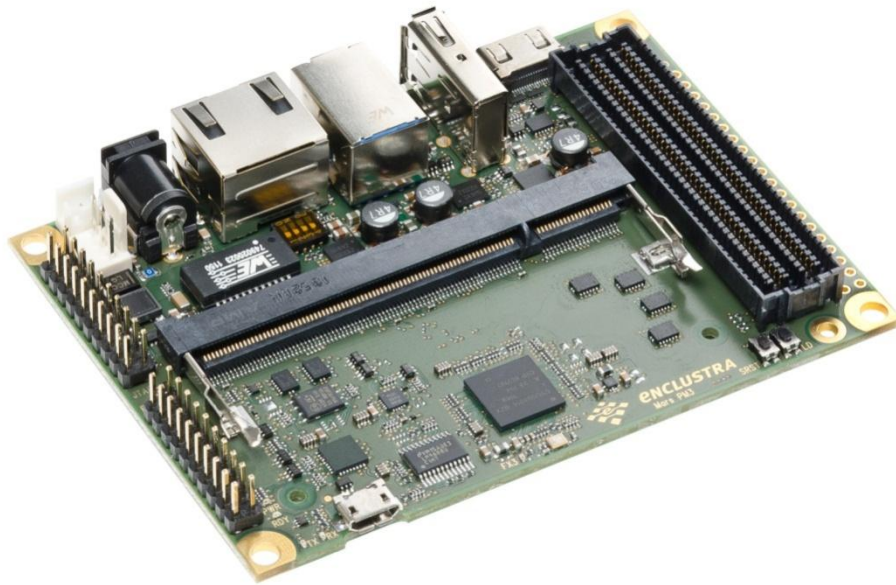
- Data and flags are valid only 2 ns before clock edge
- Three-cycle latency from ADDR to DATA/FLAGS need to be taken into account

- Careful layout is required to ensure signal integrity with 32 data lines and 100 MHz interface frequency.
- Proper timing analysis needs to be done as data and flags are valid only 2 ns before the rising edge of the clock.
- Maximum data transfer is only achievable when using auto DMA mode combined with overlapped transaction on the host side.
- The 256 kB DMA memory can only buffer data for 750 us at 330 MB/s transfer rate, which is insufficient considering the non-real-time nature of the host computer. If data loss is not acceptable, additional buffering with external memory needs to be considered.
- Achievable transfer rate is still very much dependent on the employed USB host controller, driver version and operating system. This should be getting better by now as hardware and drivers are getting more mature.
- When using 32 Bit slave fifo mode, some peripherals like SPI and UART cannot be used anymore.

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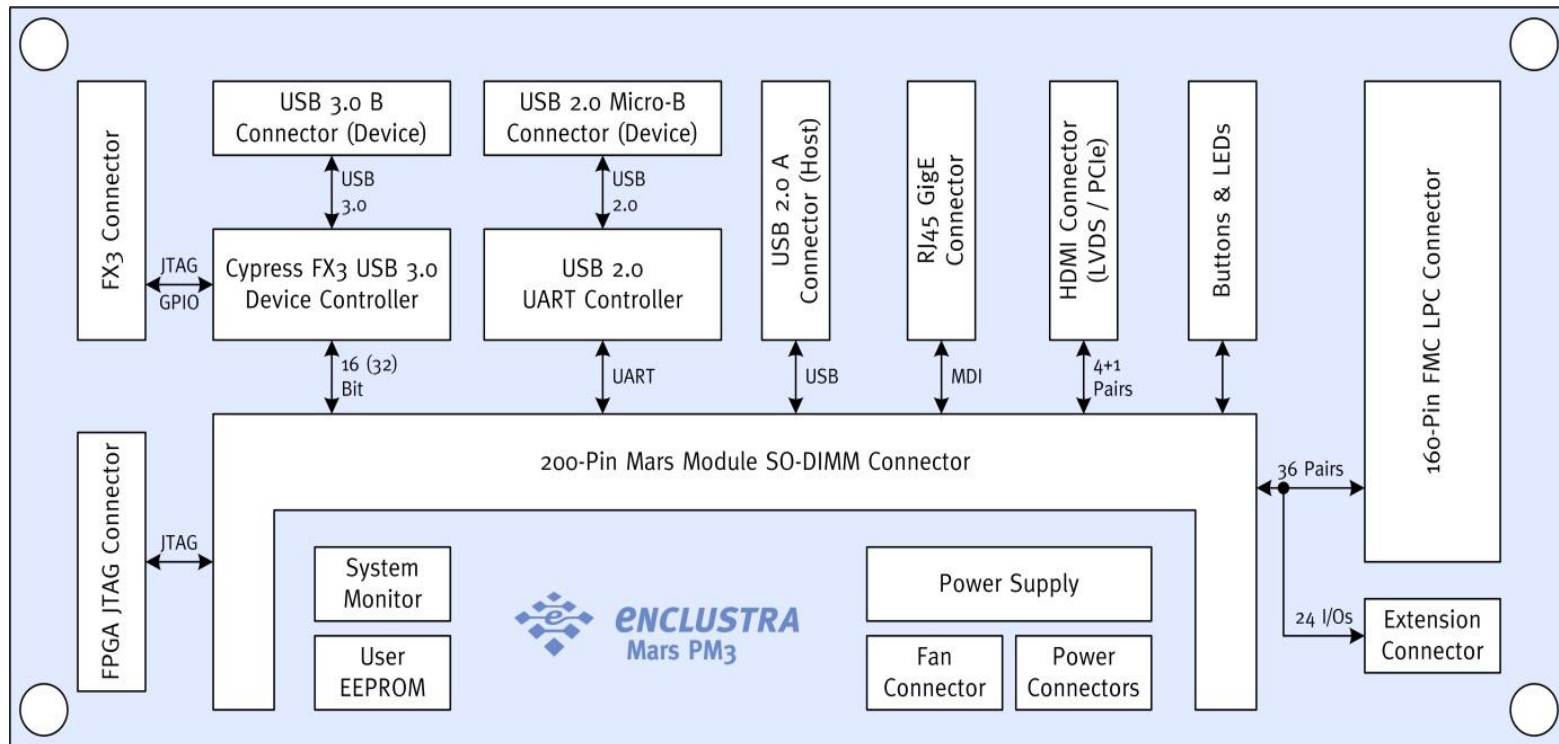


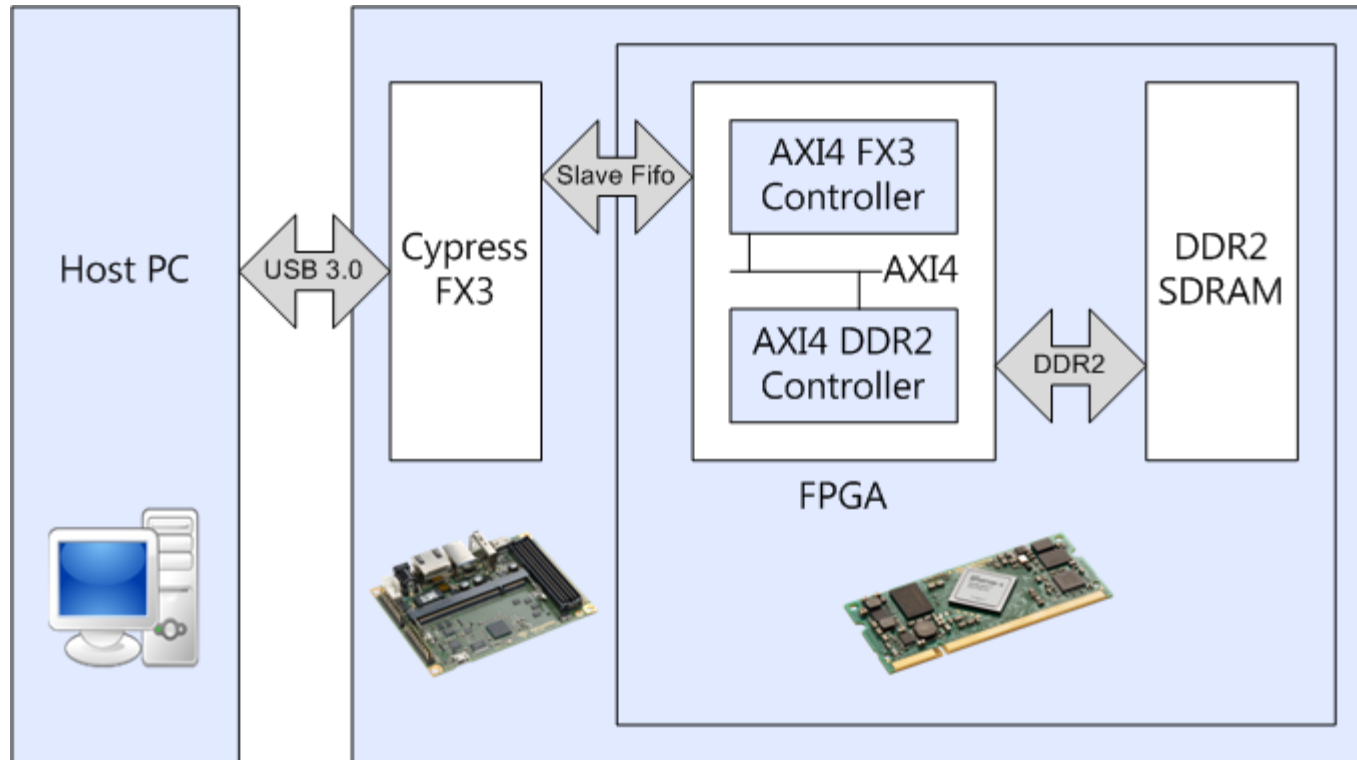
- FPGA manager allows simple data transfer between host computer and AXI bus.
- Also enables access to I2C and SPI interfaces by high-level functions



- Designed for Mars family FPGA and EP modules
- 300+ MB/sec data transfer rate over USB 3.0
- Suitable for prototype and series production
- Small solution size (p-ITX, 100x72 mm)
- Available in commercial and industrial temperature grade

Product Number	Supported Mars FPGA Modules	Temperature Range
MA-PM3-C	MA-MX1, MA-MX2, MA-AX3*, MA-ZX3*, MA-CA4*	0..+70°C
MA-PM3-I	MA-MX1, MA-MX2, MA-AX3*, MA-ZX3*, MA-CA4*	-25..+85°C





Enclustra FX3 Performance Demonstration

Setup

☐ Loopback
☐ Out Only
☒ In Only

OUT Endpoint: 0x01
IN Endpoint: 0x81

Stop on Error: ☐

Transfer Size [kB]: 4096
Start Value / Seed: 0
Send Data Pattern: Constant Byte

Device Select: Refresh
FX3

Execute

Successful Transfers: 746
Failed Transfers: 0

Reset Device Stop

Transferring data at 320.0 MB/s

Program FPGA Flash
 Program FPGA
 Program FX3 Flash
 Program FX3
 PM3 Production Test

Set Slave Fifo Mode

☒ 16 Bit
☐ 32 Bit

Test Memory

☒ FPGA Flash
☐ FX3 Flash
☐ FX3 EEPROM
☐ FMC EEPROM

Length: 4096
Address: 0

Read Voltages

I2C Access

☒ Read
☐ Write

Device: 47
Register: 0
Value: 0
16 Bit Address: ☐

Questions? Next Events



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Newsletter: subscribe@enclustra.com

Slides in PDF format:
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Next events:

- Sindex Bern



- Embedded World
26. - 28. Februar 2013
Messezentrum Nürnberg



- ¹ "AN75432 - USB 3.0 EZ-USB FX3 Orientation", Application Note, Cypress Semiconductor, www.cypress.com
- ² "USB 3.0 Sales Pitch", Presentation, Cypress Semiconductor, www.cypress.com
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